

Panel Data Models: IV-Based Methods

ECMT 676 (Econometric II) Lecture Notes

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February 24, 2014

Panel Data Models with Endogenous Explanatory Variables

- In the panel data case, where some explanatory variables are contemporaneously endogenous.
- The sampling environment is the same as before. We assume random sampling of units from a well-defined population.
- Generally, the analysis is similar to estimation by SOLS and SGLS, but we must distinguish explanatory variables from instruments.

- Write a panel data model as

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + u_{it} \quad ((1.10))$$

$$E(\mathbf{z}_{it}'u_{it}) = 0, \quad t = 1, \dots, T, \quad ((1.11))$$

where $\mathbf{x}_{it}$ is $1 \times K$. In the general case, the dimension of the instruments $\mathbf{z}_{it}$ can change with t , so $1 \times L_t$. (We will see specific examples with unobserved effects models.)

- In many cases, elements of $\mathbf{z}_{it}$ might be correlated with the errors in other time periods, so care is used in choosing $\mathbf{z}_{it}$ for each t .)

EXAMPLE: Estimating a Passenger Demand Function. Suppose passenger demand for flights is given by

$$\begin{aligned}\log(\text{passen}_{it}) &= \theta_t + \beta_1 \log(\text{fare}_{it}) \\ &\quad + \beta_2 \log(\text{dist}_i) + \beta_3 [\log(\text{dist}_i)]^2 + u_{it}\end{aligned}\quad ((1.13))$$

where $\text{lfare}_{it} = \log(\text{fare}_{it})$ is generally correlated with u_{it} .

If fares are generally higher with a higher concentration ratio, use concen_{it} as an instrument for lfare_{it} . Then

$$\mathbf{z}_{it} = (1, d2_t, \dots, dT_t, \text{concen}_{it}, \text{ldist}_i, \text{ldist}_i^2) \quad ((1.14))$$

which recognizes that a fully set of period dummies has been included.

- Suppose that $lfare_{it}$ is partially correlated with past $concen_{it}$, and

$$E(u_{it} | concen_{it}, concen_{i,t-1}, \dots, concen_{i1}, ldist_i) = 0. \quad ((1.15))$$

Then we could take

$$\mathbf{z}_{it} = (1, d2_t, \dots, dT_t, concen_{it}, concen_{i,t-1}, \dots, concen_{i1}, dist_i, ldist_i^2), \quad ((1.16))$$

which has dimension increasing in t .

- How can we efficiently use (1.15)?

- panel data model

$$\mathbf{y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{u}_i \quad ((1.17))$$

where $\mathbf{y}_i$ is $T \times 1$ and $\mathbf{X}_i$ is $T \times K$, and $\boldsymbol{\beta}$ is the $K \times 1$ vector of parameters to be estimated.

- The instrument matrix that gives us the moment conditions $E(\mathbf{z}'_{it} u_{it}) = \mathbf{0}$:

$$\mathbf{Z}'_i \mathbf{u}_i = \begin{pmatrix} \mathbf{z}'_{i1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{z}'_{i2} & \mathbf{0} & \vdots \\ \mathbf{0} & \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \dots & \mathbf{0} & \mathbf{z}'_{iT} \end{pmatrix} \begin{pmatrix} u_{i1} \\ u_{i2} \\ \vdots \\ u_{iT} \end{pmatrix} = \begin{pmatrix} \mathbf{z}'_{i1} u_{i1} \\ \mathbf{z}'_{i2} u_{i2} \\ \vdots \\ \mathbf{z}'_{iT} u_{iT} \end{pmatrix}$$

and so $E(\mathbf{Z}'_i \mathbf{u}_i) = \mathbf{0}$ is the same as

$$E(\mathbf{z}'_{it} u_{it}) = \mathbf{0}, \quad t = 1, 2, \dots, T.$$

- When the dimension of the instruments is the same for all t , so $L_t = L$, another choice is possible:

$$\mathbf{z}_i = \begin{pmatrix} \mathbf{z}_{i1} \\ \mathbf{z}_{i2} \\ \vdots \\ \mathbf{z}_{iT} \end{pmatrix}. \quad ((1.20))$$

- The IV matrix in (1.20) uses only a linear combination of the original moment conditions, namely,

$$E(\mathbf{Z}'_i \mathbf{u}_i) = \sum_{t=1}^T E(\mathbf{z}'_{it} u_{it}) = \mathbf{0}.$$

- If used properly, the instrument matrix in (1.19) will give an asymptotically more efficient estimator than that in (1.20). (But using many extra moment conditions can cause the finite-sample properties to be worse.)

The System IV Estimator

- Write the system as in the system OLS/GLS case:

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{u}_i.$$

Assumption SIV.1 (Moment Conditions): For a $T \times L$ matrix $\mathbf{Z}_i$,

$$E(\mathbf{Z}_i'\mathbf{u}_i) = \mathbf{0}. \quad \square \quad ((2.1))$$

- Assumption SIV.1 is weaker, often in important ways, than the assumption that all elements of $\mathbf{Z}_i$ are uncorrelated with all elements in $\mathbf{u}_i$: $E(\mathbf{Z}_i \otimes \mathbf{u}_i) = \mathbf{0}$.

Assumption SIV.2 (Rank Condition):

$$\text{rank } E(\mathbf{Z}_i' \mathbf{X}_i) = K. \quad \square \quad ((2.2))$$

- Looks like the key rank condition for 2SLS except that $\mathbf{Z}_i$ and $\mathbf{X}_i$ are matrices.
- A necessary condition for SIV.2 to hold is the order condition, $L \geq K$.

- If

$$\mathbf{z}_i = \begin{pmatrix} \mathbf{z}_{i1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{z}_{i2} & \mathbf{0} & \vdots \\ \mathbf{0} & \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \dots & \mathbf{0} & \mathbf{z}_{iT} \end{pmatrix} \quad ((2.4))$$

then

$$E(\mathbf{z}'_i \mathbf{x}_i) = \begin{pmatrix} E(\mathbf{z}'_{i1} \mathbf{x}_{i1}) \\ E(\mathbf{z}'_{i2} \mathbf{x}_{i2}) \\ \vdots \\ E(\mathbf{z}'_{iT} \mathbf{x}_{iT}) \end{pmatrix}. \quad ((2.5))$$

- If

$$\mathbf{z}_i = \begin{pmatrix} \mathbf{z}_{i1} \\ \mathbf{z}_{i2} \\ \vdots \\ \mathbf{z}_{iT} \end{pmatrix} \quad ((2.6))$$

then

$$E(\mathbf{z}'_i \mathbf{x}_i) = \sum_{t=1}^T E(\mathbf{z}'_{it} \mathbf{x}_{it}). \quad ((2.7))$$

- Estimation proceeds from the method of moments. The moment condition in Assumption SIV.1 can be written as

$$E[\mathbf{Z}_i'(\mathbf{y}_i - \mathbf{X}_i\boldsymbol{\beta})] = \mathbf{0} \quad ((2.7))$$

or

$$E(\mathbf{Z}_i'\mathbf{X}_i)\boldsymbol{\beta} = E(\mathbf{Z}_i'\mathbf{y}_i). \quad ((2.8))$$

- If the rank condition is violated (for example, if $L < K$), the system $E(\mathbf{Z}_i'\mathbf{X}_i)\mathbf{b} = E(\mathbf{Z}_i'\mathbf{y}_i)$ has more than one solution for $\mathbf{b}$, and $\boldsymbol{\beta}$ is not identified.

- Suppose $L = K$. Then

$$\beta = [E(\mathbf{Z}'_i \mathbf{X}_i)]^{-1} E(\mathbf{Z}'_i \mathbf{y}_i) \quad ((2.9))$$

- From here, there is only one thing to do: replace population averages with sample averages to get the **system instrumental variables (SIV)** estimator:

$$\hat{\beta}_{SIV} = \left(N^{-1} \sum_{i=1}^N \mathbf{z}'_i \mathbf{x}_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N \mathbf{z}'_i \mathbf{y}_i \right). \quad ((2.10))$$

Consistency is immediate by the usual WLLN argument.

- Alternative, we can write

$$\hat{\beta}_{SIV} = \beta + \left(N^{-1} \sum_{i=1}^N \mathbf{z}_i' \mathbf{x}_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N \mathbf{z}_i' \mathbf{u}_i \right) \quad ((2.11))$$

and use $N^{-1} \sum_{i=1}^N \mathbf{z}_i' \mathbf{u}_i \xrightarrow{p} E(\mathbf{z}_i' \mathbf{u}_i) = \mathbf{0}$.

- Equation (2.11) is more convenient for studying the asymptotic distribution.

- The SIV estimator for the panel data system with IVs stacked as in (2.6) with $L = K$ is a pooled IV estimator:

$$\hat{\beta}_{PIV} = \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{z}'_{it} \mathbf{x}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{z}'_{it} y_{it} \right). \quad ((2.12))$$

- Inference is covered generally in the next section.

Pooled 2SLS

- In the panel data model with $\mathbf{Z}_i = (\mathbf{z}'_{i1}, \dots, \mathbf{z}'_{iT})'$, the **P2SLS** estimator

$$\hat{\beta}_{P2SLS} = \left[\left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{x}'_{it} \mathbf{z}_{it} \right) \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{z}'_{it} \mathbf{z}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{z}'_{it} \mathbf{x}_{it} \right) \right]^{-1} \cdot \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{x}'_{it} \mathbf{z}_{it} \right) \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{z}'_{it} \mathbf{z}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{z}'_{it} \mathbf{y}_{it} \right)$$

- Assumption SIV.3 is equivalent to

$$\text{rank } E(\mathbf{Z}'_i \mathbf{Z}_i) = L, \quad ((3.13))$$

which extends the single equation assumption, S2SLS.2(a), in a natural way.

- When $\mathbf{Z}_j$ is block diagonal as in (2.4), the system 2SLS estimator is different. It can be obtained as follows.

(1) For each t , estimate a reduced form from a cross section regression

$$\mathbf{x}_{it} \text{ on } \mathbf{z}_{it}, \quad i = 1, \dots, N, \quad ((3.15))$$

and obtain the fitted values,

$$\mathbf{x}_{it} = \mathbf{z}_{it} \hat{\Pi}_t, \quad ((3.16))$$

where $\hat{\Pi}_t$ is $L_t \times K$ in general. Note that $\mathbf{x}_{it}$ is always $1 \times K$.

(2) Estimate β via pooled IV in the equation

$$y_{it} = \mathbf{x}_{it} \beta + u_{it}, \quad t = 1, \dots, T; \quad i = 1, \dots, N \quad ((3.17))$$

using $\mathbf{x}_{it}$ as instruments (not regressors!).

$$\hat{\beta}_{S2SLS} = \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{x}'_{it} \mathbf{x}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{x}'_{it} y_{it} \right) \quad ((3.18))$$

- The pooled 2SLS estimator estimates a common $\hat{\Pi}$ by pooling the regression in (3.15) across t (which is not possible here, since $\mathbf{z}_{it}$ has different dimensions across t).

Estimation Under Sequential Exogeneity

- We now consider IV estimation of the model

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + c_i + u_{it}, \quad t = 1, \dots, T,$$

under the sequential exogeneity assumption

$$E(u_{it} | \mathbf{x}_{it}, \mathbf{x}_{i,t-1}, \dots, \mathbf{x}_{i1}, c_i) = 0, \quad t = 1, \dots, T.$$

- Combined, the two equations imply

$$E(y_{it} | \mathbf{x}_{it}, \mathbf{x}_{i,t-1}, \dots, \mathbf{x}_{i1}, c_i) = E(y_{it} | \mathbf{x}_{it}, c_i) = \mathbf{x}_{it}\boldsymbol{\beta} + c_i$$

- A weaker form of sequential exogeneity:

$$\text{Cov}(\mathbf{x}_{is}, u_{it}) = 0, \text{ all } s \leq t.$$

- This condition suffices for consistency and is the basis for GMM moment conditions.

- After differencing to remove c_i we have many moment conditions:

$$\Delta y_{it} = \Delta \mathbf{x}_{it} \boldsymbol{\beta} + \Delta u_{it}, \quad t = 2, \dots, T,$$

$$E(\mathbf{x}'_{is} \Delta u_{it}) = E[(\mathbf{x}'_{is} (u_{it} - u_{i,t-1}))] = \mathbf{0}, \quad s = 1, \dots, t-1; \quad t = 2, \dots, T.$$

- Therefore, at time t , the available instruments in the FD equation are in the vector

$$\mathbf{x}_{i,t-1}^o \equiv (\mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{i,t-1}), \quad t = 2, \dots, T.$$

- The matrix of instruments is

$$\mathbf{Z}_i = \begin{pmatrix} \mathbf{x}_{i1}^o & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{x}_{i2}^o & \ddots & \vdots \\ \vdots & \ddots & \ddots & \mathbf{0} \\ \mathbf{0} & \cdots & \mathbf{0} & \mathbf{x}_{i,T-1}^o \end{pmatrix},$$

which has $T - 1$ rows.